

UNIVERSITY OF GUELPH · BPCH CO-OP

Marcus, here's **your plan.**

A simple, semester-by-semester course map to get you from first year to a high-paying job in the nuclear energy industry — and onto the management track from there.

Goal: nuclear chemistry → management.

Start Fall 2026 · Graduate Fall 2030 · 4 paid work terms along the way.

The plan, in plain English

You picked the nuclear path because it pays the best out of school and has the clearest ladder into management. Everything below is built to get you there.

EARN DURING SCHOOL

\$50–80K

Across your 4 paid work terms

STARTING SALARY

\$80–95K

As a station chemist, right after graduation

BY AGE ~30

\$110K+

Senior / supervisor, plus a pension

THE WHOLE PLAN IN 4 SENTENCES

1. Survive first-year physics (you'll have two physics courses — get help early, because your grades there decide whether you stay in co-op).
2. Years 2–3 mostly run themselves — the program picks your chemistry courses for you.
3. Starting in Year 2, aim every elective at nuclear: inorganic chemistry, environmental chemistry, instrumental analysis.
4. Quietly stack a few business courses (Econ, Psych, Management) so you walk into your employer already looking like a future manager.

The one thing to watch: physics

You didn't take physics in high school, so you'll take the **easier "life sciences" physics track** — that's actually good news, it's the friendlier one. Two courses, both in first year, and then **you're done with physics forever**:

- ✓ **PHYS*1300** (Fall, Year 1) — the physics course made for students who didn't take it in high school.
- ✓ **PHYS*1080** (Winter, Year 1) — the follow-up. After this, no more physics, ever.

△ WHY THIS MATTERS MORE THAN ANYTHING ELSE

To stay in the co-op program, you need a **70% average after your second semester** — which is exactly when both physics courses finish. If physics drags your average down, you could lose co-op and the whole earnings plan with it. **Get a tutor / join a study group in week one, not week six.** Treat physics like a part-time job in first year.

Your course map, year by year

Green rows are paid work terms. Everything else is school. The "nuclear picks" and "management picks" are the courses you choose on purpose.

Black = required by the program · **Blue** = nuclear-focused choice · **Orange** = management course

WHEN	COURSES	WHAT IT DOES FOR YOU
Year 1 Fall '26	CHEM*1040, BIOL*1090, MATH*1200 PHYS*1300 ECON*1050 (Microeconomics)	First-year science basics + the catch-up physics course. Econ is your first management course — learning how industries and money work.
Year 1 Winter '27	CHEM*1050, MATH*1210, BIOL*1070 PHYS*1080 + COOP*1100 PSYC*1000 (Psychology)	Finish physics. COOP*1100 preps you for landing a work term. Psych is an easy, high-mark course and useful for managing people later.
Year 1 Summer '27	Off. Get any chemistry-ish summer job and skim ahead in Organic Chemistry so Year 2 doesn't blindside you.	
Year 2 Fall '27	CHEM*2060, CHEM*2400, CHEM*2700, STAT*2040	The hardest semester — all locked in, no choices. CHEM*2400 (Analytical) is the one that lands your first lab job. Buckle down.
Year 2 Winter '28	WORK TERM 1 — a government or university lab. Builds your résumé for the bigger nuclear jobs next.	
Year 2 Summer '28	CHEM*2480, BIOC*2580, MICR*2420 BUS*2220 (Organizational Behaviour)	This is where you commit to nuclear. CHEM*2480 (spectroscopy) is core lab skill. BUS*2220 keeps building the management story.
Year 3 Fall '28	BIOC*3570, CHEM*3430 CHEM*3640 (Inorganic), ENV5*3030 (Environmental Chem)	Your most important elective choice. Inorganic + environmental chemistry are exactly what nuclear stations do every day. This makes your résumé jump out.
Year 3 Winter '29	CHEM*3750 CHEM*3650 (Inorganic II), CHEM*4020 (Instrumental Analysis) MGMT*3020 (Decision Making)	More nuclear-relevant chemistry, plus your 4th management course. By now your transcript clearly says "future leader."
Year 3 Summer '29	WORK TERM 2 — aim for OPG, Bruce Power, or Chalk River. Get them to keep you for the fall too.	
Year 4 Fall '29	WORK TERM 3 — same employer, back-to-back with WT2. This 8-month stretch is your golden opportunity: own a real project.	

WHEN	COURSES	WHAT IT DOES FOR YOU
Year 4 Winter '30	CHEM*4380 (Advanced Inorganic) MGMT*4000 (Strategic Management) + electives	The top-level nuclear chemistry course + the capstone management course. This combo is rare and powerful.
Year 4 Summer '30	WORK TERM 4 — go back and turn it into a full-time job offer before you even graduate.	
Year 5 Fall '30	Final 4000-level chemistry + electives	Light final semester. Ideally you already have a job offer in hand — finish strong and graduate.

Why we made these choices

A few decisions shaped this whole plan. Here's the short reasoning, so it's not a black box.

- ✓ **Why nuclear, not pharma or research?** Nuclear pays the most straight out of school, has a real promotion ladder into management, and comes with a pension that's worth roughly another 25% on top of salary. Pharma pays less to start; research means 5–6 more years of school. Nuclear fits your goals best.
- ✓ **Why the "easy" physics?** You didn't take high-school physics, which means you skip the brutal calculus-based physics and take the friendlier life-sciences version. Two courses, first year only, then never again.
- ✓ **Why load up on inorganic + environmental chemistry?** That's literally what nuclear station chemists do — handling metals, monitoring water chemistry, running lab instruments. These courses make employers see you as already trained for the job.
- ✓ **Why the business courses?** You want to manage. Almost no chemistry student takes Econ, Psychology, and Management on purpose. You will — and it makes you stand out for promotion the moment you're hired.
- ✓ **Why skip the research thesis?** A research project is for people going to grad school. You're going straight to industry, so we use that time on lighter courses that protect your grades instead.

THE ONE RULE TO REMEMBER

First year is about **protecting your grades through physics** so you stay in co-op. After that, every choice points at one target: **walk into a nuclear employer looking like the most prepared — and most promotable — chemist they've seen.**

Important: University of Guelph occasionally moves courses between semesters, so check this plan with a BPCH faculty advisor (bscweb@uoguelph.ca) before you register each year. Course names and salary figures verified against Guelph's official program calendar and co-op salary guide. The big-picture strategy stays the same even if a course shifts a semester.